

Use-Case Flow Specification

Group Travel Expense Simplification Engine

Use Case: View Settlement Plan

Primary Actors: Group Traveler, Trip Admin

Related Requirements: FR-001, FR-003, FR-005 (see Requirements Table)

Includes: «include» Compute Debt Settlement Graph, «include» Convert Currency

Preconditions

- The trip exists and has at least one Group Traveler member besides the requester.
- At least one expense has been recorded for the trip (see “Add Group Expense”).
- All recorded expenses have already been converted to the trip's base currency.

Postconditions

- Success: The requester is shown every member's net balance and the minimum set of suggested payments needed to settle the trip.
- Failure: No settlement plan is shown, and the requester is told why (e.g., no expenses exist yet).

Main Success Scenario

1. A Group Traveler or Trip Admin selects “View Settlement Plan” for a trip.
2. The system retrieves every recorded expense for the trip and confirms each has a base-currency amount.
3. The system calculates each member's net balance (total paid minus total owed).
4. The system invokes “Compute Debt Settlement Graph”, which runs a min-flow algorithm to find the minimum-transaction settlement (FR-001).
5. The system displays (a) each member's net balance and (b) the minimum list of suggested payments - who pays whom, and how much - to bring the group to zero balance.
6. The requester reviews the settlement plan on screen.
7. The requester may optionally choose “Export Expense Report” to download a copy («extend»).
8. The use case ends successfully.

Alternate Flow - 2a. No Expenses Recorded

1. The system finds that no expenses have been recorded for the trip.
 - 2a1. The system displays “No expenses have been added to this trip yet” and links to “Add Group Expense.”
 - 2a2. The use case ends without generating a settlement plan.